Ph.D. Course work Curriculum and Syllabi

Approved by Departmental Research Committee



Department of Information Technology
Gauhati University

Ph.D Course work Structure

Paper Code	Paper Name	Duration	Credit
IT-701	RESEARCH METHODOLOGY	1 Semester	6
IT-707	MACHINE LEARNING	1 Semester	6
IT-708	RESEARCH AND PUBLICATION ETHICS	1 Semester	2
IT-703	SEMINAR	1 Semester	6
IT-70X	SELECTED TOPICS IN CONCERNED DOMAIN OF STUDIES	1 Semester	6

IT-70X : Domain Papers

This paper aims at building a thorough understanding of the domain of research in which the Ph.D. scholar will work for, and write the thesis. The basic objective of this course is to bridge the gap between his prior academic works and activities, and the strong-foundation required for carrying the research work in the particular domain. The course work will ensure building complete foundation regarding the in-depth knowledge and skills in the particular field of study.

This paper concentrates on the very pin pointed field of studies related to the scholar's actual domain of research, and covers the topics related to the theories and practices in the domain, including the current developments.

The paper will include:

- Theories of the concerned domain
- Auxiliary knowledge and information requirement for research in the concerned domain
- Technologies and applications in the domain
- Computational techniques related to the concerned research in the domain
- Trends of research in the domain
- Current happenings in the domain
- Futuristic approaches in the domain

Paper Code	Paper Name	Credit
IT-702	COMPUTER SYSTEM	6
IT-704	DIGITAL IMAGE PROCESSING	6
IT-705	SPEECH PROCESSING	6
IT-706	NATURAL LANGUAGE PROCESSING	6
IT-709	CYBER SECURITY	6
IT-710	ARTIFICIAL INTELLIGENCE AND INTELLIGENT SYSTEMS	6
IT-711	NATURE INSPIRED COMPUTING	6
IT-712	FUZZY LOGIC AND NEURAL NETWORKS	6
IT-713	BRAIN-COMPUTER INTERFACING	6
IT-714	CLOUD COMPUTING	6

Detail Syllabus

IT- 701: Research Methodology

Credit: 6

Introduction to Computer Science Research: What is Research? Types of Research, Why Research, Significance & Status of Research in Computer Science.

Steps in Research:
Major Journals & Publications in Computer Science, Major Research Areas of Computer Science, Identification, selection & Formulation of research Problem, Developing a research Proposal, Planning your research, The wider Community, Resources & Tools, How engineering and technological research differs from scientific research.

Simulation and tools: What is Simulation? Time and randomness in simulation, Application of simulations, How a simulation model work, tools for simulations, packages for model developments.

Research Data: What is data, Mathematical, statistical and computer science views on data analysis. Statistical and mathematical theories and techniques related to Computing.

Literature Survey: Finding out about your research area, Literature search strategy, Writing critical reviews, Identifying venues for publishing your research.

Writing Papers and the Review Process: Preparing and presenting your paper. The conference review process, Making use of the referees' reports, The journal review process, Group exercise in reviewing research papers.

Thesis Writing: Planning the thesis, Writing the thesis, Thesis structure, Writing up schedule, The Oral examination(Viva Voce).

Ethical Issues and Professional Conduct: Ethics in general, Professional Ethics, Ethical Issues that Arise from Computer Technology, General Moral Imperatives, More Specific Professional Responsibilities, Organizational Leadership Imperatives.

REFERENCES:

1. **The Computer Science and Engineering (Editor-in-Chief)** *By Allen B.Tucker, JR. CRC Press, A CRC Handbook Published in Co-operation with ACM, The Association for Computing.*
2. Other standard literatures available in the internet

IT- 707: Machine Learning

Credit: 6

Unit-1: Introduction: History of machine learning, Basic concepts

Unit-2:Supervised learning: Supervised learning setup, LMS, Logistic regression, Perceptron, Backpropagation, Learning rules, Activation functions,neural networks, Exponential family, Generative learning algorithms, Gaussian discriminant analysis, Naive Bayes, Support vector machines, Model selection and feature selection, Ensemble methods: Bagging, boosting.

Unit-3:Learning theory: Bias/variance trade-off, Union and Chernoff/Hoeffding bounds, VC dimension, Worst case (online) learning.

Unit-4:Unsupervised learning: Clustering K-means, EM. Mixture of Gaussians, Factor analysis, PCA (Principal components analysis), ICA (Independent components analysis).

Unit-5:Miscellaneous topics: Hypothesis testing, cross-validation, quadratic discriminant Analysis, adaptive hierarchical clustering, gradient boosting

Unit-6: Sensitivity analysis: F1 score, Type 1 error, type 2 error, Bootsraping, Accuracies ,ROC curves, AUC and other quality metrics

Unit-7 : Hybrid Systems: Fuzzy Logic, Genetic Learning, Associative Memories ,SOM etc.

References:

1. Standard books as prescribed by the DRC from time to time on recommendation of the research guides
2. IEEE, ACM and other standard Journals

IT- 708: Research and Publication Ethics

Credit: 2

Unit-I: Research Ethics and Scientific Conduct
Philosophical Orientation to Ethics, Defining Ethics
Unit-II: Understanding Plagiarism
Conceptual understanding and early signs of plagiarism, Plagiarism: Some Cases in India, What is not Plagiarism?
Unit-III: Types of Plagiarism
Forms of Plagiarism, Plagiarism Detection Software
Unit-IV: Publication Misconduct and Publication Ethics
Publication Misconducts, FFP: Falsification, Fabrication and Plagiarism, Authorship
Unit-V: Ways to avoid Plagiarism
Plagiarism Avoiding Techniques, Paraphrasing, Steps of Paraphrasing
Unit-VI: Regulations on Plagiarism in India
Role of the students in maintaining Academic Integrity, Role of teachers/ supervisors

IT- 703: Seminar

Credit: 6

The research scholar has to prepare a seminar paper on the proposed topic, and present in two phases, mid term and final term within the semester.

The content of the seminar paper should mainly include the literature survey, and current happenings and developments in the field of study. The seminar paper will include all the road maps of the scholars Ph.D. work, including the work schedule, and visualization of the findings.

IT- 704: Digital Image Processing

Credit: 6

Introduction: Definition, origin and use of Digital Image Processing.

Digital Image Fundamentals: Elements of Visual Perception, Image Sensing, Acquisition, Sampling, Quantization.

Image enhancement in the Spatial Domain; Image enhancement in the Frequency Domain.

Image restoration; color image processing.

Wavelets and Multiresolution processing.

Image Compression.

Image Segmentation.

Recommended Readings:

1. Rafel C. Gonzalez, Richard E. Woods; Digital Image Processing, 2/e; Pearson Education.
2. Nick Efford; Digital Image Processing: a practical introduction using Java; Addison Wesley/Pearson Education.
3. Douglas A. Lyon; Image Processing in Java; PHI.

IT- 705: Speech Processing

Credit: 6

1. Signals: Continuous-time, Discrete-time and digital signals, classification of signals, Energy and power signals, Causal and Non-causal signals, Even and Odd signals.

2. Continuous-time and Discrete-time systems, Static and Dynamic system, Linear and Non-Linear systems, Time invariant and time-variant systems, Stable and unstable systems.

3. Fourier series analysis of Continuous-time Periodic signals, Fourier Transform, Dirac Function, sampling theorem, Anti aliasing filter, signal reconstruction, sampling of BP signals, Z-transform, IIR and FIR filters.

4. Anatomy and physiology of speech Production, spectrographic analysis of speech, categorization of speech sounds, speech modeling, speech complexity, Cepstral and Met cepstral measures, LPC analysis.

5. Intonation and Annotation speech: CART model, Fujisaki's model, Tilt model, KIM-model, ToBI model. Prosoty and Prosoty Boundary: Fujisaki's syntactic phrasing and prosodic phrasing.

Word Boundary: Word segmentation, Daniel-Swingly's model, PARSER model, Bayesian model, Harri's Model, Brent's generative model, Bigram word segmentation, n-gram segmentation, Chinese Restaurant Process(CRP).

6. Speech Coding

Quantization-uniform, Non uniform, Differential quantization, classification of coders, PCM, DPCM, ADPCM, DM parametric or codecs, vocoder, Mixed Excited Linear Predfiction(MELP), Hybrid coders, code Excited LINEAR prediction(CELP), sub-band coding,wide-band speech coders, speech coding standard's: ITU,GSM,MPEG, IETF, TIA, DoD, RCR(japan),ETSI, speech enhancement, speech and speaker recognition.

BOOKS:

1. Speech Signal processing- Thomas F. Quatieri, Pearson Education(Singapore)

2. Speech Production, analysis and coding- P.H. Talukdar, Lamberf Academic Publishing(Germany)

3. Signals and Systems- P. Ramesh Babu, R. Ananda Natarajan, Scitech Publication pvt. Ltd(India)

IT- 706: Natural Language Processing

Credit: 6

Introduction to Natural Language Understanding: The study of languages; Applications; evaluating language understanding systems; the different levels of language analysis; Representations, Understanding, and Organization of Systems.

Syntactic Processing: Linguistic Background; Grammars and Parsing; Features and Augmented Grammars; Grammars for Natural Language; Efficient Parsing; Ambiguity Resolution.

Semantic Interpretation: Semantics and Logical Form; Linking Syntax and Semantics; Ambiguity Resolution.

Context and World Knowledge: Knowledge representation and Reasoning; Using World Knowledge; Discourse Structure; Conversational Agents.

Recommended Readings:

1. James Allen; Natural Language Understanding 2/e; Pearson education.
2. Christopher D. Manning, Hinrich Schiitze; Foundations of Statistical Natural Language Processing; Cambridge, MA: The MIT Press.
3. Daniel Jurafsky, Jame H Martin, James H Martin; Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition; Prentice Hall.

IT-709: Cyber Security

Credit: 6

Overview: Services, Mechanisms and Attacks; The OSI security architecture; Security Models.

Classical Encryption: Symmetric Cipher Model; Substitution Techniques; Block Ciphers and the Data Encryption Standard; Differential and Linear Cryptanalysis; Block Cipher Design Principles; Block Cipher modes of Operation.

Finite Fields: Review of Groups, Rings, Fields and Modular Arithmetic.

Confidentiality using Symmetric Encryption.

Review of Number Theory.

Public Key Cryptography: Principles of Public Key Encryption; the RSA algorithm; Key Management.

Message Authentication and Hash Functions: Authentication Requirements, Functions, Message Authentication Codes, Hash Functions, Hash Algorithms.

Digital Signature and Authentication Protocols: Digital Signatures; Authentication Protocols; Digital Signature Standard.

Authentication Applications: Kerberos.

Email Security: PGP, S/MIME.

IP Security: Overview, IPSec architecture.

Web Security: SSL and Transport Layer Security; HTTPS.

System Security: Intrusion Detection; Malicious Software- Threats and Countermeasures; Firewalls- Design Principles.

Block chain Technology, Application of Block chain

Cyber Awareness, Cyber Ethics, Cyber Law

Recommended Readings:

1. William Stallings; Cryptography and Network Security- Principles and Practice, 3/e; Pearson Education.
2. Atul Kahate; Cryptography and Network Security- 2/e; Tata McGraw Hills.
3. Behrouz A. Forouzan; Cryptography & Network Security; Tata McGraw Hills.
4. Alfred J. Menezes, Paul C. van Oorschot and Scott A. Vanstone; Handbook of Applied Cryptography; CRC Press.

IT- 712: Fuzzy Logic and Neural Networks

Credit: 6

Unit –I: Introduction to Neural Networks, Introduction, Humans and Computers, Organization of the Brain, Biological Neuron, Biological and Artificial Neuron Models, Hodgkin-Huxley Neuron Model, Integrate-and-Fire Neuron Model, Spiking Neuron Model, Characteristics of ANN, McCulloch-Pitts Model, Historical Developments, Potential Applications of ANN.

Unit-II: Essentials of Artificial Neural Networks, Artificial Neuron Model, Operations of Artificial Neuron, Types of Neuron Activation Function, ANN Architectures, Classification Taxonomy of ANN –Connectivity, Neural Dynamics (Activation and Synaptic), Learning Strategy (Supervised, Unsupervised, Reinforcement), Learning Rules, Types of Application

Unit–III: Single Layer Feed Forward Neural Networks Introduction, Perceptron Models: Discrete, Continuous and Multi-Category, Training Algorithms: Discrete and Continuous Perceptron Networks, Perceptron Convergence theorem, Limitations of the Perceptron Model, Applications.

Unit-IV: Multilayer Feed forward Neural Networks Credit Assignment Problem,Generalized Delta Rule, Derivation of Backpropagation (BP) Training, Summary of Backpropagation Algorithm, Kolmogorov Theorem, Learning Difficulties and Improvements.

Unit V: Paradigms of Associative Memory, Pattern Mathematics, Hebbian Learning, General Concepts of Associative Memory (Associative Matrix, Association Rules, Hamming Distance, The Linear Associator, Matrix Memories, Content Addressable Memory), Bidirectional Associative Memory (BAM) Architecture, BAM Training Algorithms: Storage and Recall Algorithm, BAM Energy Function, Proof of BAM Stability Theorem Architecture of Hopfield Network: Discrete and Continuous versions, Storage and Recall Algorithm, Stability Analysis, Capacity of the Hopfield Network Summary and Discussion of Instance/Memory Based Learning Algorithms, Applications.

Unit –VI: Classical & Fuzzy Sets :Introduction to classical sets -properties, Operations and relations; Fuzzy sets, Membership, Uncertainty, Operations, properties, fuzzy relations, cardinalities, membership functions.

UNIT VII: Fuzzy Logic System Components: Fuzzification, Membership value assignment, development of rule base and decision making system, Defuzzification to crisp sets, Defuzzification methods.

UNIT VIII: Applications: Neural network applications: Process identification, control, fault diagnosis and load forecasting.Fuzzy logic applications: Fuzzy logic control and Fuzzy classification.

REFERENCE BOOKS:

1.Neural Networks –James A Freeman and Davis Skapura, Pearson Education, 2002.
2.Neural Networks–Simon Hakins , Pearson Education
3.Neural Engineering by C.Eliasmith and CH.Anderson, PHI
4.Neural Networks and Fuzzy Logic System by Bart Kosko, PHI Pub

IT- 713: Brain-Computer Interfacing

Credit: 6

UNIT I INTRODUCTION TO BCI

Introduction - Brain structure and function, Brain Computer Interface Types - Synchronous and Asynchronous -Invasive BCI -Partially Invasive BCI - Non Invasive BCI, Structure of BCI System, BCI Monitoring Hardware, EEG, ECoG, MEG, fMRI

UNIT II BRAIN ACTIVATION

Brain activation patterns - Spikes, Oscillatory potential and ERD, slow cortical potentials, Movement related potentials-Mu rhythms, motor imagery, Stimulus related potentials - Visual Evoked Potentials – P300 and Auditory Evoked Potentials, Potentials related to cognitive tasks.

UNIT III FEATURE EXTRACTION METHODS

Data Processing – Spike sorting, Frequency domain analysis, Wavelet analysis, Time domain analysis, Spatial filtering -Principal Component Analysis (PCA), Independent Component Analysis (ICA), Artifacts reduction, Feature Extraction - Phase synchronization and coherence.

UNIT IV MACHINE LEARNING METHODS FOR BCI

Classification techniques –Binary classification, Ensemble classification, Multiclass Classification, Evaluation of classification performance, Regression - Linear, Polynomial, RBF's, Perceptron's, Multilayer neural networks, Support vector machine, Graph theoretical functional connectivity analysis

UNIT V APPLICATIONS OF BCI

Case Studies - Invasive BCIs: decoding and tracking arm (hand) position, controlling prosthetic devices such as orthotic hands, Cursor and robotic control using multi electrode array implant, Cortical control of muscles via functional electrical stimulation. Noninvasive BCIs: P300 Mind Speller, Visual cognitive BCI, Emotion detection, Ethics of Brain Computer Interfacing

TOTAL: 45 PERIODS

Text Books

1. Rajesh.P.N.Rao, "Brain-Computer Interfacing: An Introduction", Cambridge University Press, First edition, 2013.
2. Jonathan Wolpaw, Elizabeth Winter Wolpaw, ―Brain Computer Interfaces: Principles and practice‖, Oxford University Press, USA, Edition 1, January 2012.

IT-711: Nature Inspired Computing

Credit: 6

UNIT I INTRODUCTION:
From Nature to Nature based Computing;
Parallelism;
Interactivity, Adaptation;
Feedback, Self-Organization;
Complexity, Optimization, Chaos and Fractals.

Unit II COMPUTING INSPIRED BY NATURE:
Evolutionary Computing;
Hill Climbing;
Simulated Annealing;
Genetics Principles, Standard Evolutionary Algorithm: Genetic Algorithm;
Genetic Programming.

UNIT III SWARM INTELLIGENCE:
Ant Colonies, Ant Foraging Behaviour, Ant Colony Optimization (ACO), Scope of ACO algorithms, Swarm Robotics, Particle Swarm Optimization (PSO), Various PSO based algorithms

UNIT IV IMMUNOCOMPUTING:
Immune System, Physiology and main components, Pattern Recognition and Binding, Immune Network Theory- Danger Theory, Evaluation Interaction- Immune Algorithms, Artificial Immune System, Artificial Immune Networks.

UNIT V COMPUTING WITH NEW NATURAL MATERIALS:
DNA Computing, Motivation, DNA Molecule, Adleman's experiment, Universal DNA Computers, PAM Model, Splicing Systems, Scope of DNA Computing, Migration from Classical to DNA Computing.

Reference Books:

1. L. N. de Castro, Fundamentals of Natural Computing: Basic Concepts, Algorithms, and Applications, (2006), CRC Press.
2. D. Floreano and C. Mattiussi, Bio-Inspired Artificial Intelligence: Theories, Methods, and Technologies, (2008), MIT Press.
3. D. Simon, Evolutionary Optimization Algorithms, (2013), Wiley.

IT-714: Cloud Computing

Credit: 6

Unit II
Architecture of a modern data centre;
Challenges and requirements for a cloud-centric data centre and how they differ from large, single-entity warehouse-scale computers;
Techniques behind automation and orchestration of resources, as well as key scheduling considerations in the cloud;
Cloud management techniques such as middleware, resource provisioning, metering, and orchestration.

Unit III
Major reasons for why virtualization is useful, especially on the cloud;
Different isolation types such as fault, resource, and security isolations provided by virtualization and utilized by the cloud;
Resource sharing as provided by virtualization;
Define virtualization and identify different virtual machine types such as process and system virtual machines;
Difference between classical OS virtual memory and system memory virtualization;
Recent developments in software defined networking and software defined storage from the cloud computing perspective.

Unit IV
Overall organization of data and storage;
Various types of data within the data taxonomy and classify different data types within the data taxonomy;
Problems of scale and management in big data;
Various storage abstractions;
Different types of file systems and their design considerations (e,g., Hadoop Distributed File System (HDFS), Ceph File System (CephFS);
Cloud security issues.

Unit V
Fundamental aspects of parallel and distributed programming models;
Different cloud programming models (MapReduce, Spark, GraphLab and Spark Streaming);
Main execution flow, scheduling and fault tolerance concepts in the MapReduce programming model.

Reference Books:

1. John W. itinghouse james, F.Ransome, “Cloud Computing Implementation, Management and Security” , CRC Press
2. Borko Furht, Armando Escalante, “Handbook of Cloud Computing”, Springer
3. Charles Badcock, “Cloud Revolution”, TMH